



**MODERN INSTITUTE OF TECHNOLOGY & RESEARCH
CENTRE**

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

**PROGRAMME: B.Tech. (Computer Science Engineering/ Artificial
Intelligence)**

--- COURSE FILE ---

Statistical Modeling and Forecasting (6AD4-03)



Affiliated to Bikaner Technical University, Bikaner

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**BIKANER TECHNICAL UNIVERSITY, BIKANER****बीकानेर तकनीकी विश्वविद्यालय, बीकानेर****OFFICE OF THE DEAN ACADEMICS**

VI Semester		
B. Tech. (Artificial Intelligence & Data Science)		
6AD4-03: Statistical Modeling and Forecasting		
Credit: 3	Max. Marks: 100 (IA:30, ETE:70)	
3L+0T+ 0P	End Term Exams: 3 Hours	
Course Objectives: As a result of successfully completing this course, students will : • Able to understand basic property of time-series data. • Able to handle seasonality and trend in time series data. • Able to use and deploy various models for time series data. • Able to select best model for time series data.		
Course Outcomes: Upon successful completion of the course, students will be able to CO-1: Discuss the challenges and their solutions for Time Series Data. CO-2: Understanding impact of seasonality in time series data. CO-3: Understand role of drift and trend. CO-4: Understand to working of various models used for time series data.		
S. No.	Contents	Hours
1	Basic Properties of time-series data: Distribution and moments, Stationarity, Autocorrelation, Heteroscedasticity, Normality Introduction of Time Series Analysis: Introduction to Time Series and Forecasting, Different types of data, Internal structures of time series. Models for time series analysis, Examples of Time series Nature and uses of forecasting, Forecasting Process, Data for forecasting, Resources for forecasting.	7
2	Statistics Background for Forecasting: Graphical Displays, Time Series Plots, Plotting Smoothed Data, Numerical Description of Time Series Data, Use of Data Transformations and Adjustments, General Approach to Time Series Modelling and Forecasting, Evaluating and Monitoring Forecasting Model Performance. Random walk model: Non-stationarity and unit-root process, Drift and Trend models	7
3	Introduction to Autoregressive models and forecasting: Autocorrelation and Partial autocorrelation, Autoregressive Moving Average (ARMA) Models , Autoregressive Integrated Moving Average (ARIMA) Models, Forecasting using ARIMA , Seasonal Data, Seasonal ARIMA Models Forecasting using Seasonal ARIMA Models Introduction to Vector Auto-regressive (VAR) models: Impulse Response Function (IRF), Error Correction Models, Co-integration, Vector ARIMA Models, Vector AR (VAR) Model Model Selection Criteria: Finding the “BEST” Model , Impulse Response Function to Study the Differences in Models Comparing Impulse Response Functions for Competing Models .	10

4	Time Series Regression Model: Introduction Least Squares Estimation in Linear Regression Models, Statistical Inference in Linear Regression, Prediction of New Observations, Model Adequacy Checking, Variable Selection Methods in Regression, Generalized and Weighted Least Squares, Regression Models for General Time Series Data, Exponential Smoothing, First order and Second order.	8
5	Multivariate Time Series Models and Forecasting: Multivariate Time Series Models and Forecasting, Multivariate Stationary Process Panel data models: Fixed-Effect and Random-Effect models Introduction to Neural Networks and Forecasting Spectral Analysis, Bayesian Methods in Forecasting, Principal Component Analysis (PCA) and Factor Analysis	10
	Suggested Books: 1. Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015) 2. Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017) 3. Chris Brooks “Introductory Econometrics for Finance,” Fourth Edition, Cambridge University Press 2019 4. Ruey S. Tsay “Analysis of Time-series data,” Third Edition, Wiley 2014 5. John Fox and Sanford Weisberg “An R Companion to Applied Regression,” Third Edition, SAGE 2018 6. Yves Croissant and Giovanni Millo “Panel Data Econometrics with R,” First Edition, Wiley 2018	

Course Objectives:

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- Able to use and deploy various models for time series data.
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Course Outcomes:

Upon successful completion of the course, students will be able to

CO-1: Discuss the challenges and their solutions for Time Series Data.

CO-2: Understanding impact of seasonality in time series data.

CO-3: Understand role of drift and trend.

CO-4: Understand to working of various models used for time series data.

Course Outcomes (COs) Mapping with Program Outcomes(POs) &PSO

PROGRAM OUTCOMES (POs)

Computer Science and Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problem.
 2. **Problem analysis:**Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering science.
 3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
 4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
 5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
 6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
 7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
 8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
 9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
 10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
 11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
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12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1: An ability to understand the concepts of fundamental and core concepts of Computer science & Engineering and to apply them to various areas for solving the problem like Programming, Database Management & Data Mining Systems, Compiler Design and Information security etc.

PSO 2: An ability to solve complex Computer science & Engineering problems, using programming languages and Simulation software like C, C++, Java, PHP and Matlab and so on along with analytical skills to arrive cost effective and appropriate solutions.

PSO 3: To develop the social and environmental awareness along with ethical responsibility to have a successful career and to sustain passion and zeal for real-world applications using optimal resources as an Entrepreneur.

CO, PO and PSO mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	1	2	--	2	1	--	--	--	--	--	--	--	1	3	1
CO2	1	1	2	2	--	--	--	--	--	--	--	--	1	--	1
CO3	1	2	2	3	--	--	--	--	--	--	--	--	1	3	1
CO4	1	1	2	2	3	--	--	--	--	--	--	--	1	--	1
CO5	1	1	2	2	2	--	--	--	--	--	--	--	1	--	1

1: strongly related, 2: moderately related and 3: weakly related

Class Time Table

Modern Institute of Technology and Research Centre Alwar, Raj. Department of Computer Science & Engineering VI Semester AI & DS (ROOM NO. - L-204) w.e.f. 01.01.2026							
Lecture	I	II	III	12:15- 12:45	IV	V	VI
Day / Time	9:15 - 10:15	10:15 - 11:15	11:15 - 12:15		12:45 - 01:45	01:45 - 2:45	02:45 - 03:45
Monday				$\mathcal{T}$ $\mathcal{Q}$ $\mathcal{N}$ $\mathcal{C}$ $\mathcal{H}$			
Tuesday		SMF PRK					
Wednesday	SMF PRK						
Thursday	SMF PRK						
Friday	SMF PRK						
Saturday	SMF PRK						

Lecture Plan

Class Name	B. Tech VI SEM AI&DS				
Name of the Subject	Statistical Modeling and Forecasting				
Subject Code	6AD4-03				
Proposed No. of Lectures	40				
Theory Marks	70				
Sessional Marks	30				
Total Marks	100				
Credits	3				
Class Starts On	1/1/2026				
Measurement Methodology	Mid Terms + Unit Tests + Assignments+ Quiz				
Teaching Aids Use	WB, LCD Projector, Models, Charts.				
Classes On	Monday to Saturday				
Unit No.	Lecture No.	Topic to be covered	References	Expected Date of Coverage	Actual Date of Coverage
I	1	Distribution and moments, Stationarity			
	2	Autocorrelation, Heteroscedasticity, Normality			
	3	Introduction to Time Series and Forecasting			
	4	Different types of data, Internal structures of time series			
	5	Models for time series analysis,			
	6	Examples of Time series Nature and uses of forecasting			
	7	Forecasting Process			
	8	Data for forecasting, Resources for forecasting			
II	9	Graphical Displays, Time Series Plots, Plotting Smoothed Data			

	10	Numerical Description of Time Series Data			
	11	Use of Data Transformations and Adjustments			
	12	General Approach to Time Series Modelling and Forecasting			
	13	Evaluating and Monitoring Forecasting Model Performance			
	14	Random walk model			
	15	Non-stationarity and unit-root process			
	16	Drift and Trend models			

III	17	Autocorrelation and Partial autocorrelation, Autoregressive Moving Average (ARMA) Models			
	18	Autoregressive Integrated Moving Average (ARIMA) Models, Forecasting using ARIMA			
	19	Seasonal Data, Seasonal ARIMA Models Forecasting using Seasonal ARIMA Models			
	20	Impulse Response Function (IRF), Error Correction Models			
	21	Co-integration, Vector ARIMA Models, Vector AR (VAR) Model			
	22	Finding the “BEST” Model			
	23	Impulse Response Function to Study the Differences in Models			
	24	Comparing Impulse Response Functions for Competing Models			

IV	25	Introduction Least Squares Estimation in Linear Regression			
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		Models			
	26	Statistical Inference in Linear Regression, Prediction of New Observations			
	27	Model Adequacy Checking			
	28	Variable Selection Methods in Regression			
	29	Generalized and Weighted Least Squares			
	30	Regression Models for General Time Series Data			
	31	Exponential Smoothing			
	32	First order and Second order			

V	33	Multivariate Time Series data			
	34	Multivariate Time Series Models and Forecasting			
	35	Multivariate Stationary Process			
	36	Fixed-Effect and Random-Effect models			
	37	Introduction to Neural Networks and Forecasting Spectral Analysis			
	38	Bayesian Methods in Forecasting			
	39	Principal Component Analysis (PCA)			
	40	Factor Analysis			

Unit-1	1	Time Series Decomposition Techniques			
Unit-2	2	Forecast Accuracy Measures			
Unit-3	3	Handling Missing Values and Outliers in Time Series Data			
Unit-4	4	State Space Models and Kalman Filter			
Unit-5	5	Real-Time Time Series			

		Forecasting Applications			
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References/Text Books	
1	Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)
2	Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)
3	Ruey S. Tsay “Analysis of Time-series data,” Third Edition, Wiley 2014

Sample Notes

UNIT-1

STATISTICAL MODELING & FORECASTING LECTURE NO: -1

Basic Properties of time-series data: Distribution and moments, Stationarity, Autocorrelation, Heteroscedasticity, Normality:

- 1. Distribution and Moments:** Time-series data may follow various distributions, such as normal, exponential, or Poisson. Moments like mean, variance, skewness, and kurtosis describe the shape and characteristics of the distribution.
- 2. Stationarity:** A time series is stationary if its statistical properties like mean, variance, and autocorrelation structure do not change over time. This implies that the series has a constant mean and variance and the covariance between two time points only depends on the time lag between them.
- 3. Autocorrelation:** Autocorrelation measures the degree of similarity between a time series and a lagged version of itself over different time lags. It helps in identifying patterns or dependencies within the data.
- 4. Heteroscedasticity:** Heteroscedasticity refers to the situation where the variability of a variable is unequal across different values of another variable or across different time points in the case of time-series data. It violates the assumption of constant variance in many statistical models.
- 5. Normality:** Normality refers to the assumption that the distribution of the residuals or errors in a statistical model follows a normal distribution. Deviations from normality can affect the validity of statistical inferences, especially in hypothesis testing and parameter estimation.

Basically there are **3 characteristics of a time series**: stationarity, trend and seasonality
Prerequisites: time series definition, statistics such as mean, variance, covariance.

An observed time series can be decomposed into three components: the trend (long term direction), the seasonal (systematic, calendar related movements) and the irregular (unsystematic, short term fluctuations)

One defining characteristic of a time series is that it is a list of observations where the ordering matters. Ordering is very important because there is dependency and changing the order could change the meaning of the data.

Basic concepts are the words that are necessary for comprehension of incoming information and performance of daily tasks. The correct understanding and usage of basic concepts is essential for effective communicative exchanges in your child's early years as well as success in academia in your child's later years.

The basic formula of Time Series Data:

$y(t) = x(t)\beta + \varepsilon(t)$, where $y(t) = \{y_t; t = 0, \pm 1, \pm 2, \dots\}$ is a sequence, indexed by the time subscript t , which is a combination of an observable signal sequence $x(t) = \{x_t\}$ and an unobservable white-noise sequence $\varepsilon(t) = \{\varepsilon_t\}$ of independently and identically distributed random variables.

We assume that the time series values we observe are the realizations of random variables $y_1, \dots, y_t$, which are in turn part of a larger stochastic process $\{y_t; t \in \mathbb{Z}\}$.

In time series analysis, the analogs to the mean and the variance are the mean function and the autocovariance function.

The mean of a series is defined as
 $\mu_t = E(y_t)$.

The autocovariance function is defined as
 $\gamma(s, t) = \text{cov}(y_s, y_t) = E[(y_s - \mu_s)(y_t - \mu_t)]$.

The autocovariance measures the linear dependence between two points (y_s, y_t) at different times. For smooth series the autocovariance function stays large even when the s and t are far apart, whereas for choppy series the autocovariance function is close to zero for large separations.

If $s=t$ it follows that

$$\gamma(t, t) = E[(y_t - \mu_t)^2] = \text{var}[y_t],$$

As in classical statistics, it is more convenient to deal with a measure of association between -1 and 1 . The autocorrelation function (ACF) is computed from the autocovariance function by dividing by the standard deviations of y_s and y_t .

$$\rho(s, t) = \gamma(s, t) / (\gamma(s, s)\gamma(t, t))$$

The autocorrelation, also called serial correlation, is a measure of the internal correlation of a time series. It is a representation of the degree of similarity between the time series and a lagged version of itself. High autocorrelation values mean that the future is strongly correlated to the past.

The cross-covariance function is a measure of predictability of one series y_t from another series x_s .

$$\gamma_{xy}(s,t) = \text{cov}(x_s, y_t) = E[(x_s - \mu_{x_s})(y_t - \mu_{y_t})]$$

The cross-covariance function can be scaled to $[1, -1]$, referred to as cross-correlation function (CCF).

$$\rho_{xy}(s,t) = \gamma_{xy}(s,t) / (\gamma_x(s,s) \gamma_y(t,t))$$

Estimation of Correlation The real values for the mean and the autocorrelation function are in general not known and must be estimated based on the sample data $y_1, y_2, \dots, y_n$.

The mean function is estimated by the sample mean

$$\bar{y} = \frac{1}{n} \sum_{t=1}^n y_t \text{ and the theoretical autocorrelation function is estimated by the sample ACF}$$

$$\hat{\rho}(k) = \hat{\gamma}(k) / \hat{\gamma}(0) = \frac{\sum_{t=1}^{n-k} (y_{t+k} - \bar{y})(y_t - \bar{y})}{\sum_{t=1}^n (y_t - \bar{y})^2}, \text{ for } k=0, 1, \dots, n-1.$$

Stationarity and non-stationarity time series A key idea in time series analysis is that of stationarity. Stationarity is considered as an important precondition for the analysis of the correlational structure in the time series. A time series is considered stationary if its behavior does not change over time. This means, for example, that the values always tend to vary about the same level and that their variability is constant over time.

A time series $\{y_t: t \in \mathbb{Z}\}$ is said to be strictly stationary if for any $k > 0$ and any $t_1, \dots, t_k \in \mathbb{Z}$, the distribution of $(y_{t_1}, \dots, y_{t_k})$ is the same as that for $(y_{t_1+u}, \dots, y_{t_k+u})$ for every value of u . Following this definition the stochastic behavior of the process does not change through time. A weaker definition of stationarity is second order stationarity, also referred to as wide-sense stationarity or covariance stationary. Here, we do not assume anything about the joint distribution of the random responses $y_1, y_2, y_3, \dots$ except that the mean is constant $E[y_t] = \mu$ and that the covariance between two observations y_t and y_{t+k} depends only on the lag k between two observations and not on the point t in the time series.

The theory for time series is based on the assumption of second-order stationarity. However, real-life data are often not stationary. The assumptions of stationarity above only applies after trends/seasonal effects have been removed.

References/Text Books

- 1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)".
- 2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"
- 3 Ruey S. Tsay "Analysis of Time-series data," Third Edition, Wiley 2014

70 Important Questions Bank

Part-A

1. What is Time Series Data?
2. What are the moments of a distribution in time series?
3. What is stationarity in time series analysis?
4. What is autocorrelation in time series?
5. What is heteroscedasticity in time series?
6. What is normality in time series data?
7. What are the components of a time series?
8. What is forecasting in time series?
9. What are different types of data used in forecasting?
10. What is the forecasting process?
11. What are graphical displays in time series analysis?
12. What is a time series plot?
13. What is smoothing in time series data?
14. What is numerical description of time series data?
15. What are data transformations in time series analysis?
16. What is the random walk model?
17. What is a unit root process?
18. What is a drift model in time series?
19. What is an autoregressive (AR) model?
20. What is the moving average (MA) model?
21. What is the ARMA model?
22. What is the ARIMA model?
23. What is seasonal ARIMA?
24. What is partial autocorrelation?
25. What is the Vector Autoregressive (VAR) model?
26. What is the Impulse Response Function (IRF)?
27. What is cointegration in time series?
28. What is an error correction model (ECM)?
29. What are model selection criteria?
30. What is time series regression?
31. What is least squares estimation?
32. What is prediction in regression models?
33. What is model adequacy checking?
34. What is exponential smoothing?
35. What is first-order exponential smoothing?
36. What is second-order exponential smoothing?
37. What are multivariate time series models?
38. What are panel data models?
39. What is the fixed-effect model?
40. What is Principal Component Analysis (PCA) in forecasting?

Part-B

1. Calculate the mean and variance of the time series data: 4, 6, 8, 10, 12.
2. Find the first difference of the time series: 12, 15, 18, 22, 25.
3. Calculate a 3-period moving average for the data: 10, 12, 14, 16, 18.
4. Using the random walk model, find (Y_5) if $(Y_4 = 20)$ and error $(\epsilon_5 = 3)$.
5. Forecast the next value using AR(1) model where $(Y_t = 0.7Y_{t-1})$ and $(Y_{t-1} = 10)$.
6. Calculate simple exponential smoothing forecast if $(Y_t = 50)$, previous forecast $(F_t = 45)$, and $(\alpha = 0.3)$.
7. Calculate covariance between two series (X) and (Y) where $(X = 2,4,6)$ and $(Y = 3,5,7)$.
8. Compute Mean Squared Error (MSE) if actual values are 20, 25, 30 and forecasts are 18, 27, 29.
9. Find autocorrelation at lag 1 for data: 3, 5, 7.
10. Forecast using ARIMA differencing for series 10, 14, 18, 22.
11. Explain the Autoregressive (AR) model in detail.
12. Explain the Moving Average (MA) model.
13. Explain the ARIMA model used in forecasting.
14. Explain exponential smoothing in time series forecasting.
15. Explain the Vector Autoregressive (VAR) model.

Part-C

1. Compute the coefficient of variation for the time series data: 20, 25, 30, 35, 40. Explain its significance.
2. Using the linear trend model $(Y_t = a + bt)$, estimate parameters if data are: Year ((t)): 1, 2, 3, 4 and Sales: 10, 14, 18, 22.
3. Compute lag-1 autocovariance for the series: 3, 6, 9, 12, 15.
4. Estimate the next value using the AR(2) model $(Y_t = 0.5Y_{t-1} + 0.3Y_{t-2})$ for dataset: 10, 12, 15.
5. Apply weighted moving average with weights 0.5, 0.3, 0.2 for the data: 40, 45, 50.
6. Compute Mean Absolute Error (MAE) for actual values: 100, 110, 120, 130 and forecasts: 98, 115, 118, 132.
7. Calculate seasonal index if quarterly sales are: Q1 = 200, Q2 = 240, Q3 = 260, Q4 = 300 and average annual sales = 250.
8. Determine if the series shows a linear trend using first differences: 5, 9, 13, 17, 21.
9. Compute covariance matrix for two variables $(X = (2,4,6))$ and $(Y = (1,3,5))$.
10. Forecast next value using exponential smoothing with $(\alpha = 0.4)$, $(Y_t = 80)$, previous forecast $(F_t = 75)$.
11. Explain the concept of stationarity and test it using dataset: 2, 4, 6, 8, 10.

12. Explain the concept of heteroscedasticity with example dataset: 2, 4, 8, 16, 32.
13. Explain the Partial Autocorrelation Function (PACF) using dataset: 5, 7, 9, 11, 13.
14. Explain the concept of Vector Autoregressive (VAR) models using two datasets: GDP = (100,110,120), Investment = (40,45,50).
15. Explain Principal Component Analysis (PCA) in time series forecasting using dataset variables (X_1, X_2, X_3).

Modern Institute of Technology & Research Centre, Alwar

Assignment-1

Branch: AI&DS Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date of Issue: 20-01-2026

Last Date of Submission: 28-01-2026

Q1. Explain the basic properties of time series data. Discuss in detail the concepts of distribution and moments, stationarity.

Q2. Define Time Series Analysis and Forecasting. Explain the different types of data, internal structures of a time series (trend, seasonal, cyclical, and irregular components), and discuss the nature and uses of forecasting.

Q3. The following time series represents the annual sales (in units) of a product:

Year	1	2	3	4	5
Sales	120	125	131	138	146

- Calculate the **mean** and **variance** of the series.
- Comment on whether the series appears to be **stationary or non-stationary**, giving reasons based on the behavior of mean and variance.

Q4.

Year	2019	2020	2021	2022	2023
Production	50	54	59	63	68

- Determine the **trend values** using the **method of semi-averages**.
- Comment on the nature of the trend.

Q5.

t	1	2	3	4	5
(X _t)	10	12	15	18	20

- Compute the lag-1 autocorrelation coefficient.
- Interpret the result and comment on the dependence between successive observations

Modern Institute of Technology & Research Centre,
Alwar Assignment-2

Branch: AI&DS

Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date of Issue: 09-02-2026

Last Date of Submission: 16-02-2026

Q1. Explain the steps involved in time series modelling and forecasting, including:

1. graphical display and identification of components
2. stationarity checking
3. model selection

Q2. Explain how log transformation, differencing, and seasonal adjustment help in improving model performance and making the series stationary.

Q3. Explain non-stationarity, unit root process, and the difference between random walk, random walk with drift, and trend models with suitable equations.

Q4. The quarterly demand data is:

120, 135, 150, 165, 180, 195

1. Compute the 3-point moving average to obtain smoothed values.
2. Plot the original and smoothed series (time series plot).

Q5. Consider the random walk with drift model:

$$Y_t = 2 + Y_{t-1} + \varepsilon_t$$

Given $Y_0 = 50$ and assuming error term $\varepsilon_t = 0$:

1. Find forecasts for Y_1, Y_2, Y_3 .
2. Determine whether the series is stationary or non-stationary.

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Assignment-3

Branch: AI&DS

Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date of Issue: 21-02-2026

Last Date of Submission: 28-02-2026

Q1. A time series follows an AR(1) model given by

$$Y_t = 0.7Y_{t-1} + \varepsilon_t$$

where ε_t is white noise with mean zero. If $Y_{10} = 50$, compute the one-step, two-step, and three-step ahead forecasts. Also compute the theoretical autocorrelation at lag 1 and lag 2.

Q2. A time series becomes stationary after first differencing and follows an ARMA(1,1) model:

$$\Delta Y_t = 0.6\Delta Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}$$

Given $\Delta Y_{20} = 5$ and $\varepsilon_{20} = 2$, compute the one-step ahead forecast for Y_{21} .

Q3. Consider quarterly seasonal data modeled by a SARIMA model:

$$SARIMA(1,1,1)(1,1,0)_4$$

- (a) Write the full model equation using backshift operator notation.
- (b) If the AIC values of three competing models are 520, 505, and 498, determine the best model using model selection criteria.
- (c) Briefly explain how Impulse Response Function (IRF) can be used to compare two competing VAR models.

Q4. Explain Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) and describe how they are used to identify AR, MA, and ARMA models.

Q5. Define Vector Auto-Regressive (VAR) models and explain the concepts of Co-integration, Error Correction Models (ECM), Vector ARIMA models, and Impulse Response Function (IRF). Discuss how model selection criteria such as AIC and BIC help in finding the best model and how IRF is used to study differences between competing VAR models.

Modern Institute of Technology & Research Centre, Alwar

Assignment-4

Branch: AI&DS

Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date of Issue: 11-03-2026

Last Date of Submission: 18-03-2026

Q1. Explain the concept of Least Squares Estimation in Linear Regression Models.

Q2. What is Model Adequacy Checking in Regression Analysis?

Q3. Least Squares Regression

The following data represents the relationship between advertising cost (X) and sales (Y).

X	Y
1	2
2	3
3	5
4	7
5	8

Tasks:

1. Fit a simple linear regression model

$$Y = a + bX$$

2. Estimate the values of a and b using the least squares method.
3. Predict the value of Y when X = 6.

Exponential Smoothing (First Order)

The monthly demand for a product is given below.

Month	Demand
1	50
2	55
3	53
4	60
5	58

Given smoothing constant $\alpha = 0.3$ and initial forecast $F_1 = 50$.

Tasks:

1. Calculate the forecast values for months 2 to 5 using first order exponential smoothing.

Formula:

$$F_{t+1} = \alpha Y_t + (1 - \alpha)F_t$$

Q5. Given a case study ,a company wants to study how advertising expenditure (X_1) and price (X_2) affect the sales (Y) of a product. The following data is collected for 5 time periods.

Time	Advertising (X_1)	Price (X_2)	Sales (Y)
1	2	10	25
2	4	8	30
3	6	7	40
4	8	6	50
5	10	5	60

Assume the multiple linear regression model

$$Y = a + b_1X_1 + b_2X_2$$

Tasks

1. Using the Least Squares Method, estimate the regression coefficients

$$a, b_1, b_2$$

2. Write the estimated regression equation.
3. Predict the sales (Y) when

$$X_1 = 12 \text{ and } X_2$$

4. Interpret the meaning of the coefficients b_1 and b_2 .

References, Journals, websites and URLs

References (Books)

1. Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. *Time Series Analysis: Forecasting and Control*. Wiley.
2. Hyndman, R. J., & Athanasopoulos, G. *Forecasting: Principles and Practice*. OTexts.
3. Montgomery, D. C., Jennings, C. L., & Kulahci, M. *Introduction to Time Series Analysis and Forecasting*. Wiley.
4. Hamilton, J. D. *Time Series Analysis*. Princeton University Press.

Journals

1. International Journal of Forecasting — Official journal of the International Institute of Forecasters. (ScienceDirect)
2. Journal of Forecasting — Publishes research on statistical forecasting and predictive modeling. (Wikipedia)
3. Forecasting (MDPI) — Open-access journal covering forecasting methods and applications. (MDPI)
4. Foresight: The International Journal of Applied Forecasting — Focuses on practical forecasting applications in business and industry. (Wikipedia)

Websites

1. OTexts – Forecasting: Principles and Practice — Free online textbook for time series forecasting.
2. International Institute of Forecasters — Professional organization for forecasting research and practice.
3. NIST Statistical Engineering Handbook — Statistical methods, regression, and forecasting concepts.
4. StatQuest with Josh Starmer — Easy explanations of statistical modeling and machine learning concepts.

URL Links

1. Forecasting Principles and Practice (Online Book)
2. International Journal of Forecasting Homepage
3. Journal of Forecasting Wiley Page
4. NIST Time Series and Forecasting Handbook

Video Lecture Links:

1. **Bayesian Methods in Econometrics and Forecasting – Lecture Series**
[ISBA Bayesian Forecasting Video Lectures](#)
Covers Bayesian econometric modeling, forecasting methods, and advanced statistical techniques. ([bayesian.org](#))
2. **Time Series Forecasting using Statistical and Machine Learning Models**
[PyData NYC Forecasting Lecture](#)
Explains VAR models, neural networks, and forecasting approaches. ([pyvideo.org](#))
3. **Applying Statistical Modeling and Machine Learning to Perform Time-Series Forecasting**
[PyData LA Forecasting Lecture](#)
Covers forecasting workflows, pitfalls, and practical applications. ([pyvideo.org](#))
4. **Practical Time Series Forecasting Video Series**
[Practical Time Series Forecasting Videos](#)
A complete series covering ARIMA, forecasting metrics, visualization, and business forecasting. ([forecastingbook.com](#))
5. **NPTEL – Time Series Analysis Lecture**
[NPTEL Time Series Analysis Lecture](#)
Covers trends, seasonality, stationarity, and correlation concepts. ([Class Central](#))
6. **Aric LaBarr – YouTube Time Series Playlist**
[Aric LaBarr Time Series Videos](#)
Includes ARIMA, Prophet, VAR, GARCH, Bayesian models, and forecasting evaluation. ([mail.ariclabarr.com](#))
7. **NPTEL – Time Series Modelling and Forecasting with Applications in R**
[NPTEL Forecasting with R Course](#)
A detailed academic course on time series processes and forecasting using R. ([terna.digimat.in](#))

5th Sem Result

B. TECH BATCH 2023 5 SEM RESULT BRANCH: AI&DS			SGPA	CGPA
S.NO.	Roll Number	Name		
1	23EMCAD001	AASHIF KHAN	8.77	0
2	23EMCAD002	ABHISHEK SINGH	9.7	8.76
3	23EMCAD003	ADITYA YADAV	7.56	0
4	23EMCAD004	AMAN AGGRWAL	7.18	7.25
5	23EMCAD005	ANKIT GURJAR	8.06	0
6	23EMCAD006	ANKIT PALIWAL	8.41	0
7	23EMCAD007	ANUJ SHARMA	9.25	8.47
8	23EMCAD008	ARJUN KUMAR	9.5	8.27
9	23EMCAD009	ARMAN HUSSAIN	9.04	8.65
10	23EMCAD010	BHAVYA KHANDELWAL	9.54	8.81
11	23EMCAD011	CHAMAN JANGID	7.04	6.74
12	23EMCAD012	CHIRAG AVASTHI	9.89	9.55
13	23EMCAD013	DEEPAK SHARMA	9.66	8.4
14	23EMCAD014	DEVENDER SINGH	8.12	0
15	23EMCAD015	DIVESH KUMAR	7.97	7.83
16	23EMCAD017	GAURAV SAXENA	8.75	8.32
17	23EMCAD018	HARSH KUMAR	9.12	7.92
18	23EMCAD019	INDERJEET SINGH	5.62	0
19	23EMCAD020	JITENDRA SINGH	8.12	7.95
20	23EMCAD021	KANAK SHUKLA	0	0
21	23EMCAD022	KASHISH OAD	9.33	8.41
22	23EMCAD023	KASHISH YOGI	0	0
23	23EMCAD024	KUNAL SHARMA	8.27	7.97
24	23EMCAD025	KUSH SHARMA	9.66	8.56
25	23EMCAD026	LAKSHYA KUMAR SAIN	8.35	7.29
26	23EMCAD027	LOKESH KUMAR	8.6	8.93
27	23EMCAD028	LOKESH KUMAR TRIVEDI	7.27	0
28	23EMCAD029	MAHIKA KHANNA	8.91	8.16
29	23EMCAD030	MANOJ SAINI	0	0
30	23EMCAD031	MS CHANCHAL NARUKA	10	9.7
31	23EMCAD032	MS ISHITA GUPTA	9.04	8.25
32	23EMCAD033	MS. AARCHI JAIN	9.7	9.21
33	23EMCAD034	MS. KOMAL VERMA	0	0
34	23EMCAD035	MS. LAVISHKA BHARDWAJ	8.33	8.14
35	23EMCAD036	MS. MANISHA BALAI	9.58	9.21
36	23EMCAD037	MS. NISHA	9.62	8.92
37	23EMCAD038	MS. RITIKA CHAWLA	8.02	7.77
38	23EMCAD039	MS. RIYA GOUD	9.35	8.73

39	23EMCAD040	MS. SHREETI KUMARI	9.02	8.81
40	23EMCAD041	MS. VANSHIKA GUPTA	9.54	9.7
41	23EMCAD042	NIKHIL SHARMA	9.62	8.64
42	23EMCAD044	PRIYANKA GUPTA	9.62	9.51
43	23EMCAD045	PRIYANSHU KHANDELWAL	9.04	9.1
44	23EMCAD046	PUNIT KUMAR	0	0
45	23EMCAD047	RISHABH SAINI	8.08	8.41
46	23EMCAD048	RISHI RAJ SONI	6.64	0
47	23EMCAD049	ROHIT GUPTA	8.39	7.74
48	23EMCAD050	SHIVANI	9.97	9.59
49	23EMCAD051	SHREEYA SANCHI	9.7	8.52
50	23EMCAD052	SUJAL MATHURIA	8.2	7.53
51	23EMCAD053	SURENDER SINGH	7.16	7.36
52	23EMCAD054	TANMAY TOMAR	7.58	0
53	23EMCAD055	TUSHAR JANGID	9.04	0
54	23EMCAD056	VANSH SHARMA	8.72	0
55	23EMCAD057	VIJAY TRIPATHI	7.25	0
56	23EMCAD058	VINEET SAINI	8.56	0
57	23EMCAD059	VINIT JAIN	8.79	0
58	23EMCAD060	VIVEK JANGID	6.93	0
59	23EMCAD063	YUNIS	0	0



B. TECH. VI SEM I MID TERM EXAMINATION 2025-26

BRANCH: AI&DS

SUB. CODE: 6AD4-03

SUBJECT: SMF

TIME ALLOWED: 2 Hours

Name of Faculty: Pradeep Kumar

MAX. MARKS: 30

PART A

Note: Attempt All Questions. (Each Question 2 Marks)

(Short Answer up to 25 words)

- Q1. Explain distributions and moments in Time Series data. (CO1)
- Q2. Explain internal structure of Time Series data. (CO1)
- Q3. Write any two data transformation techniques used to transform Time Series data. (CO2)
- Q4. Explain the techniques used to smooth the data. (CO2)
- Q5. What does (1,1,1) represents in ARIMA(1, 1, 1) model. (CO3)

PART B

Note. Attempt any Four Questions. (Each Question 3 Marks)

(Analytical/ Problem Solving questions)

- Q1. Differentiate between stationary and non-stationary time series data. (CO1)
- Q2. Explain the components of Time Series data. Also explain additive and multiplicative model. (CO1)
- Q3 Explain non-stationarity and unit root process.(CO2)
- Q4. Consider the time series: 10, 12, 14, 16, 18. Compute the 3-period moving average. (CO2)
- Q5. A random walk model is given by: (CO2)

$$Y_t = Y_{t-1} + e_t$$

If $Y_0 = 50$ and errors are 2, -1, 3, find Y_1, Y_2, Y_3 .

- Q6. For ARMA(1,1) given: (CO3)

$$Y_t = 0.7Y_{t-1} + e_t + 0.5e_{t-1}$$

If $Y_0 = 5$, $e_1 = 1$, $e_0 = 0$, find Y_1 .

PART C

Note. Attempt any two Questions. (Each Question 4 Marks)

(Descriptive/ Analytical/ Problem Solving/ Design Questions)

Q1. Discuss the complete forecasting process. Also explain the uses of forecasting in real-life. (CO1)

Q2. Given the time series data: (CO2)

5, 7, 9, 11, 13, 15

- (a) Compute the 3-period moving average.
- (b) Determine whether the series shows a trend.

Q3. Given VAR(1) model: (CO3)

$$Y_{1t} = 0.5Y_{1t-1} + 0.3Y_{2t-1}$$

$$Y_{2t} = 0.2Y_{1t-1} + 0.4Y_{2t-1}$$

If $Y_{1,0} = 2$, $Y_{2,0} = 3$, find $Y_{1,1}$ and $Y_{2,1}$.



B. TECH. VI SEM II MID TERM EXAMINATION 2025-26

BRANCH: AI&DS

SUB. CODE: 6AD4-03

SUBJECT: SMF

TIME ALLOWED: 2 Hours

Name of Faculty: Pradeep Kumar

MAX. MARKS: 30

PART A

Note: Attempt All Questions. (Each Question 2 Marks)

(Short Answer up to 25 words)

- Q1. What is meant by a Time Series Regression Model? (CO4)
Q2. Define Least Squares Estimation in Linear Regression. (CO4)
Q3. What is Exponential Smoothing? Write the purpose of using it in time series forecasting. (CO4)
Q4. What is the difference between Fixed-Effect and Random-Effect models in panel data? (CO5)
Q5. What is the role of Principal Component Analysis in multivariate data analysis? (CO5)

PART B

Note. Attempt any Four Questions. (Each Question 3 Marks)

(Analytical/ Problem Solving questions)

- Q1.** Explain the steps involved in building a regression model for time series data. (CO4)
Q2. What are variable selection methods in regression? Briefly explain any two methods. (CO4)
Q3 What is a multivariate time series? How is it different from a univariate time series? (CO5)
Q4. A time series has the following observations: (CO4)

Time	Actual Value
1	20
2	24
3	30

Using first-order exponential smoothing with smoothing constant $\alpha = 0.5$ and initial forecast $F_1 = 20$, find the forecast for time period F4.

- Q5.** For the following data: (CO4)

X	Y
2	5
4	9
6	13

Fit a simple linear regression model $Y = a + bX$ using least squares method and find the value of slope b .

- Q6.** A dataset has three variables with eigenvalues: (CO5)

$$\lambda_1 = 5, \lambda_2 = 3, \lambda_3 = 2$$

Using Principal Component Analysis, calculate the proportion of total variance explained by the first principal component.

PART C

Note. Attempt any two Questions. (Each Question 4 Marks)

(Descriptive/ Analytical/ Problem Solving/ Design Questions)

Q1. Discuss Bayesian methods used in time series forecasting. (CO5)

Q2. Explain how neural networks can capture nonlinear patterns in time series data and compare them briefly with traditional statistical forecasting methods. (CO5)

Q3. Consider the following time series data: (CO4)

Time (t)	Y
1	30
2	34
3	40
4	46

Using Holt's (second-order) exponential smoothing method with $\alpha = 0.5$ (level smoothing constant) and $\beta = 0.4$ (trend smoothing constant), and initial values:

$$L_1 = 30, T_1 = 4$$

where L_t is the level component and T_t is the trend component.

1. Compute the level L_t and trend T_t values up to $t = 4$.
2. Calculate the forecast for time period $t = 5$.

MTT-1, 2 Marks

S.NO.	ROLL NO.	STUDENT NAME	I MTT	II MTT
1	23EMCAD001	AASHIF KHAN	11	10
2	23EMCAD002	ABHISHEK SINGH	10	10
3	23EMCAD003	ADITYA YADAV	0	2
4	23EMCAD004	AMAN AGGRWAL	3	8
5	23EMCAD005	ANKIT GURJAR	4	4
6	23EMCAD006	ANKIT PALIWAL	1	6
7	23EMCAD007	ANUJ SHARMA	21	ABSENT
8	23EMCAD008	ARJUN KUMAR	ABSENT	17
9	23EMCAD009	ARMAN HUSSAIN	17	18
10	23EMCAD010	BHAVYA KHANDELWAL	23	1
11	23EMCAD011	CHAMAN JANGID	ABSENT	0
12	23EMCAD012	CHIRAG AVASTHI	25	24
13	23EMCAD013	DEEPAK SHARMA	20	ABSENT
14	23EMCAD014	DEVENDER SINGH	12	17
15	23EMCAD015	DIVESH KUMAR	7	7
16	23EMCAD016	GAURAV	FREEZE	FREEZE
17	23EMCAD017	GAURAV SAXENA	ABSENT	2
18	23EMCAD018	HARSH KUMAR	9	2
19	23EMCAD019	INDERJEET SINGH	0	0
20	23EMCAD020	JITENDRA SINGH	0	0
21	23EMCAD021	KANAK SHUKLA	8	11
22	23EMCAD022	KASHISH OAD	26	24

23	23EMCAD023	KASHISH YOGI	3	ABSENT
24	23EMCAD024	KUNAL SHARMA	6	14
25	23EMCAD025	KUSH SHARMA	13	15
26	23EMCAD026	LAKSHYA KUMAR SAIN	0	3
27	23EMCAD027	LOKESH KUMAR	20	19
28	23EMCAD028	LOKESH KUMAR TRIVEDI	25	4
29	23EMCAD029	MAHIKA KHANNA	1	14
30	23EMCAD030	MANOJ SAINI	0	0
31	23EMCAD031	MS CHANCHAL NARUKA	25	3
32	23EMCAD032	MS ISHITA GUPTA	24	12
33	23EMCAD033	MS. AARCHI JAIN	28	21
34	23EMCAD034	MS. KOMAL VERMA	22	9
35	23EMCAD035	MS. LAVISHKA BHARDWAJ	16	14
36	23EMCAD036	MS. MANISHA BALAI	25	18
37	23EMCAD037	MS. NISHA	23	14
38	23EMCAD038	MS. RITIKA CHAWLA	16	12
39	23EMCAD039	MS. RIYA GOUD	27	12
40	23EMCAD040	MS. SHREETI KUMARI	25	26
41	23EMCAD041	MS. VANSHIKA GUPTA	ABSENT	14
42	23EMCAD042	NIKHIL SHARMA	18	17
43	23EMCAD043	PRAMOD KUMAR	FREEZE	FREEZE
44	23EMCAD044	MS. PRIYANKA GUPTA	19	14
45	23EMCAD045	PRIYANSHU KHANDELWAL	24	19
46	23EMCAD046	PUNIT KUMAR	12	8
47	23EMCAD047	RISHABH SAINI	DEBARRED	9
48	23EMCAD048	RISHI RAJ SONI	8	12
49	23EMCAD049	ROHIT GUPTA	ABSENT	7

50	23EMCAD050	SHIVANI	29	25
51	23EMCAD051	SHREEYA SANCHI	19	ABSENT
52	23EMCAD052	SUJAL MATHURIA	8	7
53	23EMCAD053	SURENDER SINGH	19	2
54	23EMCAD054	TANMAY TOMAR	1	6
55	23EMCAD055	TUSHAR JANGID	29	13
56	23EMCAD056	VANSH SHARMA	28	8
57	23EMCAD057	VIJAY TRIPATHI	19	7
58	23EMCAD058	VINEET SAINI	13	6
59	23EMCAD059	VINIT JAIN	11	2
60	23EMCAD060	VIVEK JANGID	16	16
61	23EMCAD061	YASH KUMAWAT	FREEZE	FREEZE
62	23EMCAD063	YUNIS	15	9

Modern Institute of Technology & Research Centre, Alwar

UnitTest-1

Branch: AI&DS Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date :

Note: Attempt any three questions.

Max Marks:30

Q1. Explain the basic properties of time-series data. Discuss distribution and moments, stationarity, autocorrelation, heteroscedasticity, and normality, highlighting their importance in time-series analysis.

Q2. Describe the introduction to time series analysis and forecasting. Explain the different types of data, the internal structure of a time series (trend, seasonal, cyclical, and irregular components), and the nature and uses of forecasting.

Q3. Given the following time-series data:

Time (t)	Value
1	12
2	15
3	14
4	18
5	20

a) Compute the mean and variance of the series.

b) Comment on whether the series appears to be stationary based on its mean behavior.

Q4. Consider the time-series data given below:

Time (t)	Sales
1	100
2	120
3	110
4	130

5	150
---	-----

a) Calculate the lag-1 autocorrelation coefficient.

b) Interpret the result and state whether the data shows positive or negative autocorrelation.

Modern Institute of Technology & Research Centre, Alwar

UnitTest-2

Branch: AI&DS Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date :

Note: Attempt any three questions.

Max Marks:30

Q1. ARMA (1,1)

$$Y_t = 0.6Y_{t-1} + 0.5\varepsilon_{t-1} + \varepsilon_t$$

Initial values:

$$Y_0 = 1.5, \varepsilon_0 = 0.2$$

Errors:

$$t = 1: \varepsilon_1 = -0.3$$

$$t = 2: \varepsilon_2 = 0.4$$

$$t = 3: \varepsilon_3 = -0.1$$

Calculate: Y_1, Y_2, Y_3

Q2. VAR(2) Model

$$Y_{1,t} = 0.5 + 0.3Y_{1,t-1} + 0.2Y_{2,t-1} + 0.1Y_{1,t-2} + 0.1Y_{2,t-2}$$

$$Y_{2,t} = 1.0 + 0.4Y_{1,t-1} + 0.3Y_{2,t-1} + 0.2Y_{1,t-2} + 0.1Y_{2,t-2}$$

t	Y ₁	Y ₂
3	2.0	3.0
4	2.8	4.0
5	3.0	5.0

Calculate: $Y_{1,5}, Y_{1,6}, Y_{2,5}, Y_{2,6}$

Also explain Impulse Response Function and its interpretation in VAR model.

Q3. ARIMA (p,d,q)

$$\phi = 0.5, \theta = 0.4$$

Month	Sales
Jan	112
Feb	118
Mar	132
Apr	129
May	121
Jun	138

Using ARIMA terms, calculate the forecasted value.

Q4.

Explain Stationary Data.

Explain the steps involved in SARIMA $(p, d, q)(P, D, Q)_s$.

6Th Sem UT-1, 2 marks:

S.NO.	ROLL NO.	STUDENT NAME	U1	U2
			MM:30	MM:30
1	23EMCAD001	AASHIF KHAN	6	16
2	23EMCAD002	ABHISHEK SINGH	9	14
3	23EMCAD003	ADITYA YADAV	5	10
4	23EMCAD004	AMAN AGGRWAL	4	0
5	23EMCAD005	ANKIT GURJAR	6	6
6	23EMCAD006	ANKIT PALIWAL	13	10
7	23EMCAD007	ANUJ SHARMA	16	9
8	23EMCAD008	ARJUN KUMAR	16	6
9	23EMCAD009	ARMAN HUSSAIN	11	15
10	23EMCAD010	BHAVYA KHANDELWAL	10	14
11	23EMCAD011	CHAMAN JANGID	12	14
12	23EMCAD012	CHIRAG AVASTHI	16	11
13	23EMCAD013	DEEPAK SHARMA	19	16
14	23EMCAD014	DEVENDER SINGH	12	16
15	23EMCAD015	DIVESH KUMAR	6	13
16	23EMCAD016	GAURAV	0	0
17	23EMCAD017	GAURAV SAXENA	12	15
18	23EMCAD018	HARSH KUMAR	8	15
19	23EMCAD019	INDERJEET SINGH	2	13
20	23EMCAD020	JITENDRA SINGH	5	12
21	23EMCAD021	KANAK SHUKLA	10	12
22	23EMCAD022	KASHISH OAD	17	16
23	23EMCAD023	KASHISH YOGI	10	11
24	23EMCAD024	KUNAL SHARMA	3	12
25	23EMCAD025	KUSH SHARMA	9	16
26	23EMCAD026	LAKSHYA KUMAR SAIN	8	12
27	23EMCAD027	LOKESH KUMAR	6	10
28	23EMCAD028	LOKESH KUMAR TRIVEDI	7	14
29	23EMCAD029	MAHIKA KHANNA	12	6
30	23EMCAD030	MANOJ SAINI	9	14
31	23EMCAD031	MS CHANCHAL NARUKA	13	14
32	23EMCAD032	MS ISHITA GUPTA	17	12
33	23EMCAD033	MS. AARCHI JAIN	25	22

34	23EMCAD034	MS. KOMAL VERMA	12	15
35	23EMCAD035	MS. LAVISHKA BHARDWAJ	8	17
36	23EMCAD036	MS. MANISHA BALAI	16	17
37	23EMCAD037	MS. NISHA	15	18
38	23EMCAD038	MS. RITIKA CHAWLA	8	12
39	23EMCAD039	MS. RIYA GOUD	14	12
40	23EMCAD040	MS. SHREETI KUMARI	15	15
41	23EMCAD041	MS. VANSHIKA GUPTA	10	14
42	23EMCAD042	NIKHIL SHARMA	13	14
43	23EMCAD043	PRAMOD KUMAR	10	11
44	23EMCAD044	MS. PRIYANKA GUPTA	10	16
45	23EMCAD045	PRIYANSHU KHANDELWAL	18	15
46	23EMCAD046	PUNIT KUMAR	9	10
47	23EMCAD047	RISHABH SAINI	7	13
48	23EMCAD048	RISHI RAJ SONI	6	16
49	23EMCAD049	ROHIT GUPTA	12	14
50	23EMCAD050	SHIVANI	15	20
51	23EMCAD051	SHREEYA SANCHI	8	12
52	23EMCAD052	SUJAL MATHURIA	8	8
53	23EMCAD053	SURENDER SINGH	10	10
54	23EMCAD054	TANMAY TOMAR	8	14
55	23EMCAD055	TUSHAR JANGID	17	25
56	23EMCAD056	VANSH SHARMA	11	8
57	23EMCAD057	VIJAY TRIPATHI	8	7
58	23EMCAD058	VINEET SAINI	13	12
59	23EMCAD059	VINIT JAIN	7	10
60	23EMCAD060	VIVEK JANGID	9	15
61	23EMCAD061	YASH KUMAWAT	0	0
62	23EMCAD063	YUNIS	10	13

Mid-Term Marks AI&DS:

VI SEM (6AD4-03)

	MTT-1	MTT-2
Below Average (<40%)	20	30
Average(>=40 & <=60)	10	18
Good(60>& <=80)	12	5
Excellent (>80)	11	2

Unit Test Performance

VI SEM (6AD4-03)

	UT-1	UT-2
Below Average Students (<40)	38	19
Average Students (>=40 and <=60)	22	40
Good Students (>60 and <=80)	1	2
Excellent Students (>80)	1	1

S. NO.	PERFORMANCE	MARKS
1	Excellent	>24
2	Good	[18-24]
3	Average	[12-18]
4	Below average	< 12

Table: Guidelines to identify slow learners

Identification Criteria	Actions taken
Students scoring less than 40% of marks in Internal Assessment	• Mentor pursues the student's growth frequently; guide them about attending classes and making up missed classes. • Intimating parents about their ward's performance. • Conduction of remedial classes

Table: Guidelines to identify Bright students

Identification Criteria	Actions taken
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Students awarded as First Class with Distinction (FCD) in their Semester exams.	• Students are motivated to take up mini projects and encouraged to participate in inter college national fests. • Encouraged to attend conferences, workshops and publish papers. • Encouraged for preparation of competitive examinations like GATE, GRE etc.
Top two students of each class	• Awarded with certificate and cash prize.
Students securing ranks at University level	• Distribution of Gold medals.

Continuous Improvement

Time Table for Zero Lectures for Weak Students

Day/Time	3:45 - 4:45
	Zero Hour
Monday	SMF- IV SEM AI&DS PK
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	

SMF-Statistical Modeling & Forecasting

PK- Pradeep Kumar

Methodology to Encourage Students

- To stimulate to make self study-Notes.

- Take notice frequently on their work.
- Instruct them in spare time as Tutor.
- Recognize their inabilities.
- To assign constantly assignments.

Motivate to solve the Numerical Problems of various Text Books